5.3.14 I/O port characteristics

General input/output characteristics

Unless otherwise specified, the parameters given in Table 48 are derived from tests performed under the conditions summarized in Table 18. All I/Os are designed as CMOS and TTL-compliant.

Note: For information on GPIO configuration, refer to the application note STM32 GPIO configuration for hardware settings and low-power consumption (AN4899).

Table 48. I/O static characteristics

The I/O structure options listed in this table can be a concatenation of options including the option explicitly listed. For instance, TT_a refers to any TT I/O with _a option. TT_xx refers to any TT I/O and FT_xx refers to any FT I/O.
All I/Os are CMOS- and TTL-compliant (no software configuration required). Their characteristics cover more than the strict CMOS-technology or TTL parameters. The coverage of these requirements is shown in the figure below.
The minimum and maximum values are specified for a junction temperature (T_J) of 125°C.

Symbol	Parameter	Condition	Min	Typ	Max	Unit
V_{IL}	I/O input low level voltage	2.7 V<V_{DDIOx}<3.6 V	-	-	$0.3V_{DD}$ (1)	V
	I/O input low level voltage		-	-	$0.4\ V_{DD} - 0.1$(2)	
V_{IH}	I/O input high level voltage	2.7 V < V_{DDIOx} < 3.6 V	$0.7V_{DD}$ (1)	-	-	V
	I/O input high level voltage		$0.52V_{DD}$ + 0.18(2)	-	-	
V_{HYS} (2)	TT_xx, FT_xxx and NRST I/O input hysteresis	2.7 V < V_{DDIOx} < 3.6 V	-	300	-	mV
I_{leak} (3)	FT_xx Input leakage current(2)	0 < V_{IN} ≤ Max(V_{DDXXX})(4)	-	-	±-200	nA
		Max(V_{DDXXX}) < V_{IN} ≤ Max(V_{DDXXX} + 1 V)(5)(6)(4)	-	-	±2500	
		Max(V_{DDXXX} + 1) < V_{IN} ≤ 5.5 V (5)(6)(4)	-	-	750	
	TT_xx Input leakage current	0 < V_{IN} ≤ Max(V_{DDXXX} + 1 V)(4)	-	-	±200	
R_{PU}	Weak pull-up equivalent resistor(7)	$V_{IN} = V_{SS}$	30	40	50	kΩ
R_{PD}	Weak pull-down equivalent resistor(7)	$V_{IN} = V_{DD}$	30	40	50	
C_{IO}	I/O pin capacitance	-	-	5	-	pF

1. *Compliant with CMOS requirements.*
2. *Specified by design - Not tested in production.*
3. *This parameter represents the pad leakage of the I/O itself. The total product pad leakage is provided by the following formula: ITotal_Ileak_max = 10 µA + [number of I/Os where V_{IN} is applied on the pad] × Ilkg max.*
4. *Max(V_{DDXXX}) is the maximum value of all the I/O supplies.*
5. *To sustain a voltage higher than the minimum of V_{DD} or V_{DDA} plus 0.3 V, disable the internal pull-up and pull-down resistors.*
6. *V_{IN} must be less than Max(V_{DDXXX}) + 3.6 V.*
7. *The pull-up and pull-down resistors are designed with a true resistance in series with a switchable PMOS/NMOS. This PMOS/NMOS contribution to the series resistance is minimal (~10% order).*

All I/Os are CMOS- and TTL-compliant (no software configuration required). Their characteristics cover more than the strict CMOS-technology or TTL parameters. The coverage of these requirements is shown in the following figure.